Assessment 4

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1.Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze

performance, security, and manageability, then justify the better choice.

Type-1 vs Type-2 Hypervisors (Cloud Datacenter)

Comparison

Factor	**Type-1 Hypervisor**	**Type-2 Hypervisor**
Deployment	Runs directly on hardware	Runs on host OS
Performance	High (no OS overhead)	Lower (extra OS layer)
Security	More secure (smaller attack surface)	Less secure (host OS vulnerabilities)
Manageability	Enterprise-level tools	Limited for large scale

Analysis

- **Performance:** Type-1 is faster because it directly interacts with hardware.
- **Security:** Type-1 is more secure due to isolation from host OS.
- **Manageability:** Type-1 supports centralized management for large data centers.

Justification

Type-1 hypervisor is the better choice for mission-critical cloud applications because it offers **higher performance, stronger security, and better scalability**.

2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

How Virtualization Supports Isolation

- Each workload runs in a **separate virtual machine (VM)**
- Hypervisor enforces **resource isolation (CPU, memory, storage)**
- Virtual networks isolate traffic between tenants

For Different Domains (Finance, Healthcare, Student)

- Separate VMs or containers per domain
- Role-based access control (RBAC)
- Network segmentation (VLANs, VPCs)

Two Major Attack Surfaces

1. **Hypervisor attacks** (VM escape, privilege escalation)
2. **Shared resource attacks** (side-channel attacks on CPU/cache)

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

Given Metrics

- CPU: 92% → Very high
- Memory: 88% → Near saturation
- Storage latency: 35 ms → High (should be low)
- VM boot time: 170 s → Slow

Likely Bottlenecks

- CPU contention → Too many VMs
- Memory pressure → Insufficient RAM
- Storage I/O bottleneck → Slow disks or overload

Corrective Actions

- Add more CPU cores or distribute workloads
- Increase RAM or optimize VM allocation
- Use SSD storage / improve IOPS

- Reduce overcommitment of resources

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

Traditional Datacenter

- Hardware-based configuration
- Manual provisioning
- Slow scaling

SDDC Improvements

Compute Virtualization

- VMs and containers enable flexible resource allocation

Storage Virtualization

- Abstracts physical storage into pools
- Enables dynamic allocation

Network Virtualization

- Software-defined networking (SDN)
- Automated routing and security policies

Conclusion

SDDC improves **agility, automation, scalability, and faster deployment** compared to traditional systems.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue andpropose a secure identity and privacy design using federation concepts.

Problem Analysis

- Different cloud providers use different identity systems
- Causes authentication mismatch and access failure

Solution: Federation-Based Identity Design

Key Concepts

- **Identity Federation** (Single identity across clouds)
- **Single Sign-On (SSO)**
- **Trust relationships between providers**

Secure Design

- Use **Identity Provider (IdP)** (e.g., centralized auth)
- Use protocols like:
  - SAML
  - OAuth 2.0
  - OpenID Connect

Privacy Measures

- Minimal data sharing
- Token-based authentication
- Encryption of identity data

Conclusion

Federation ensures **secure, seamless access across multiple clouds while maintaining privacy and consistency**.